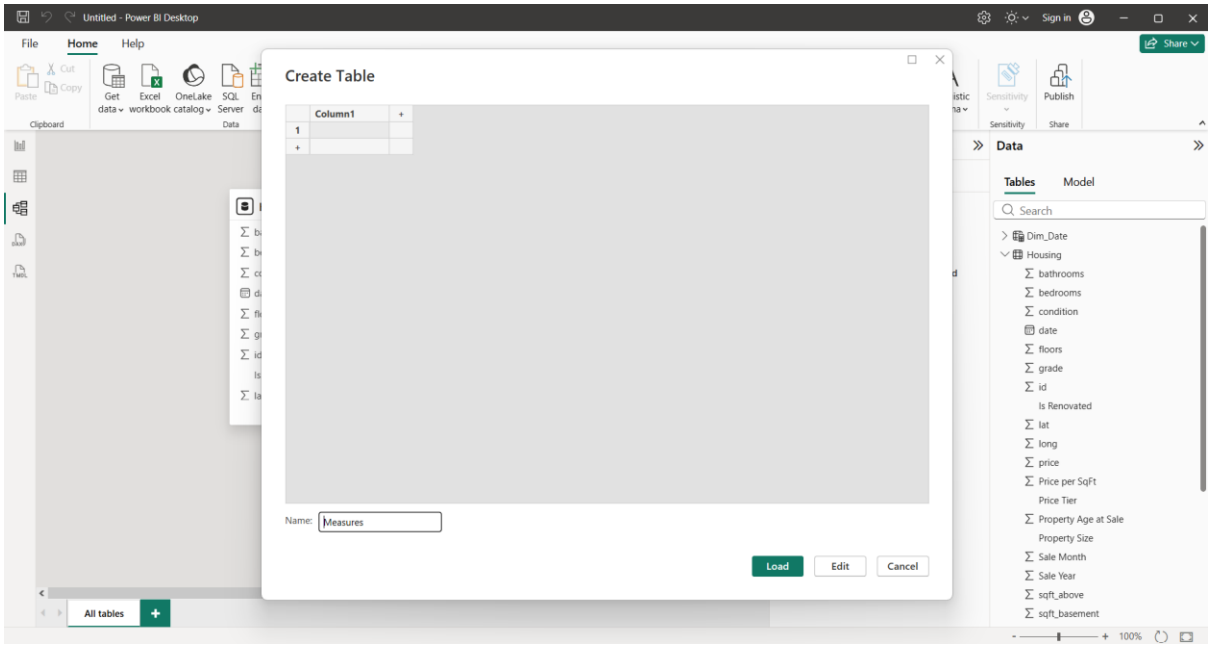
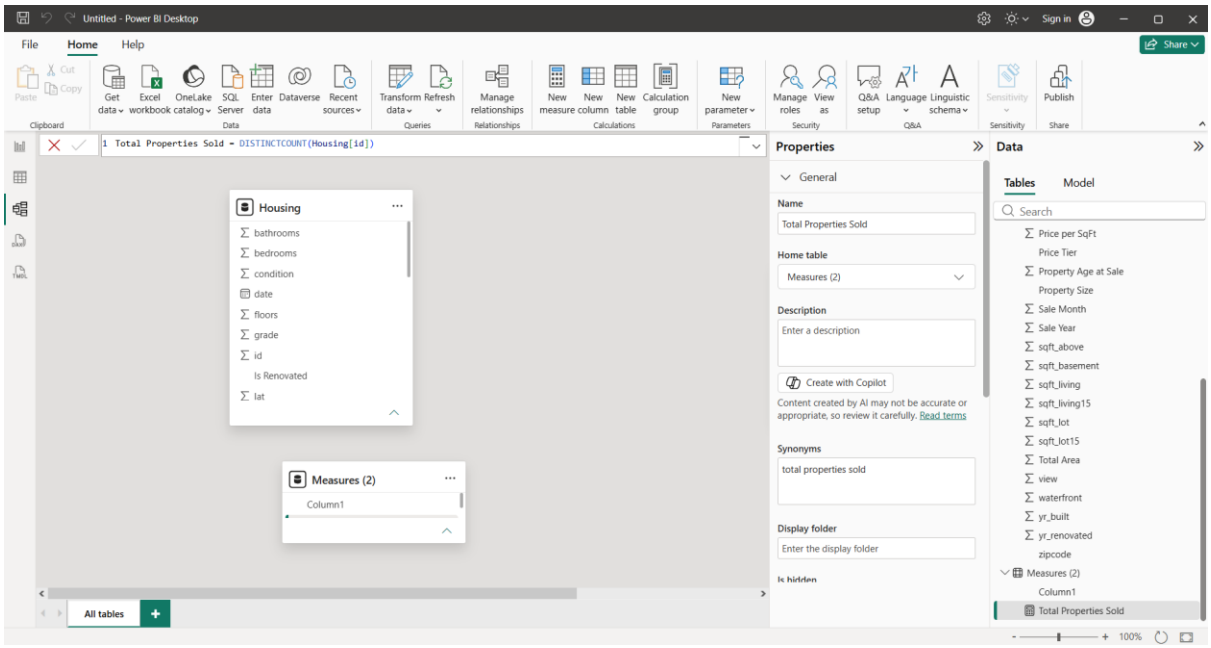
WEEK 10

SAI SHRUTI B



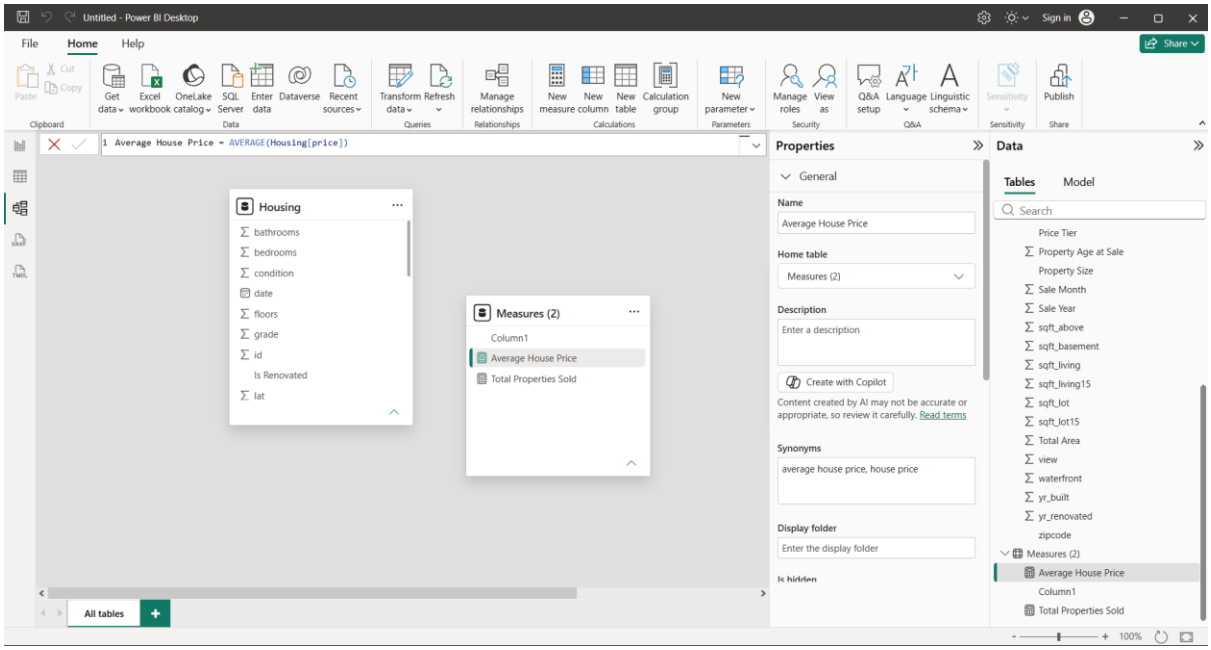
A separate table named Measures was created in Power BI using the Enter Data option. This table provides a dedicated location for organizing the DAX measures required for the real-estate analytics dashboard.



The Total Properties Sold measure was created using the DAX formula:

Total Properties Sold = DISTINCTCOUNT(Housing[id])

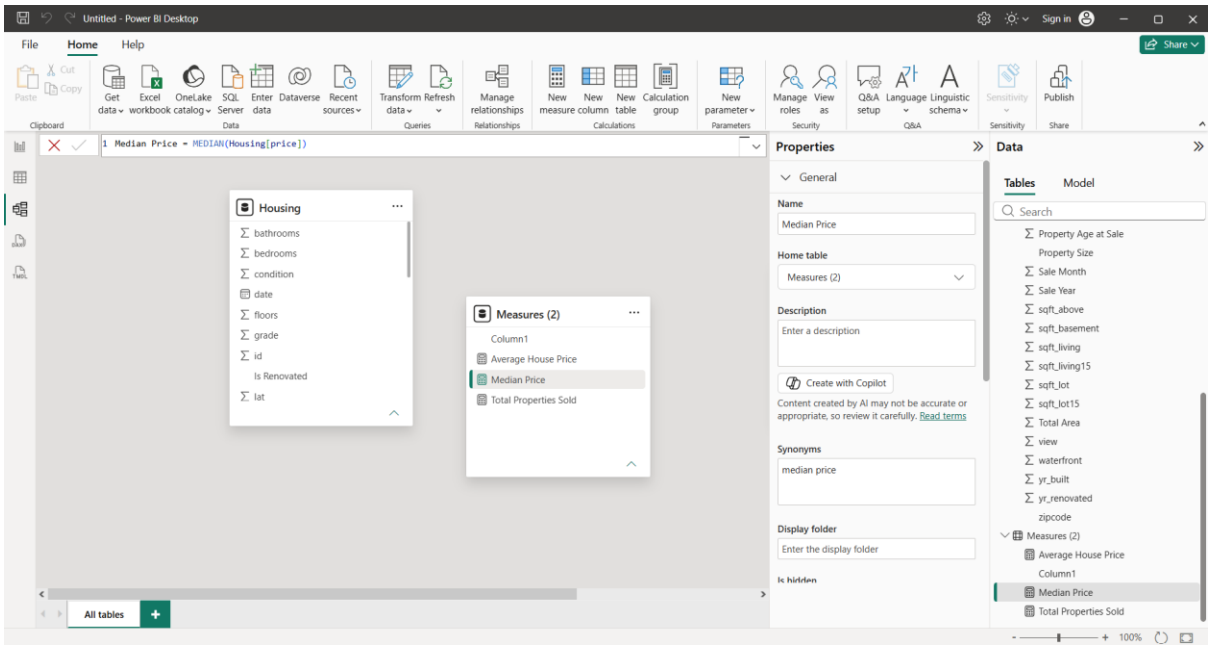
This measure counts the distinct property IDs in the Housing table and provides the total number of properties represented in the dataset.



The Average House Price measure was created using the AVERAGE function:

Average House Price = AVERAGE(Housing[price])

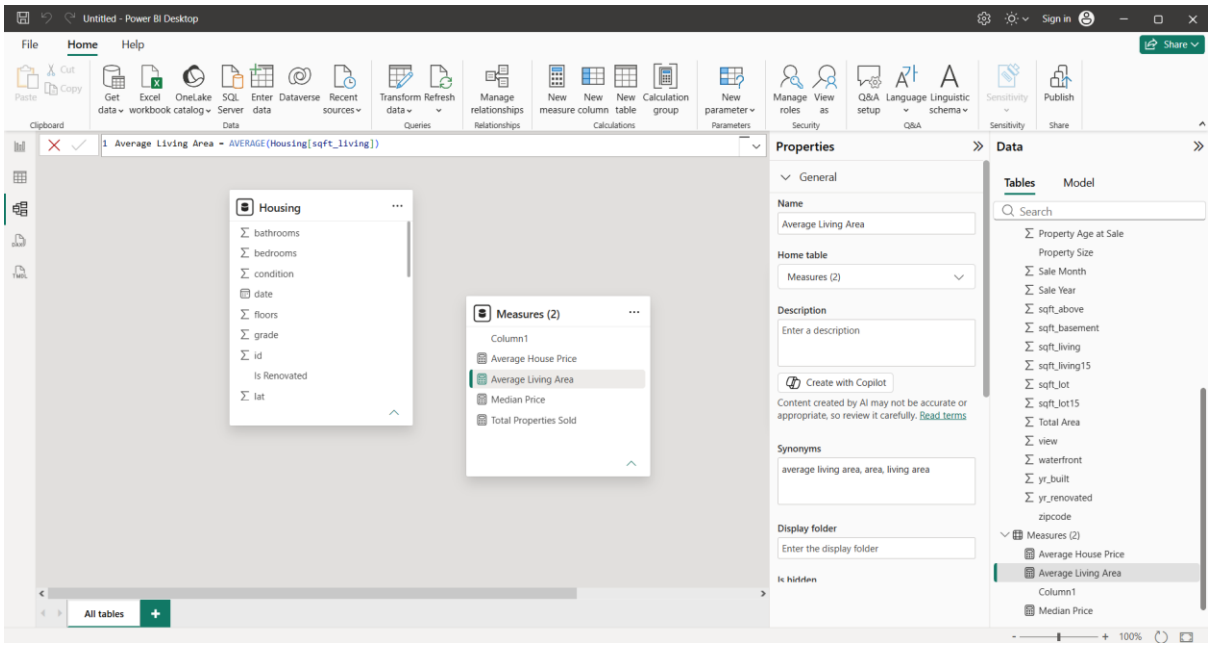
This measure calculates the average sale price of properties in the dataset and can be used to summarize house prices in the dashboard.



The Median Price measure was created using the MEDIAN function:

Median Price = MEDIAN(Housing[price])

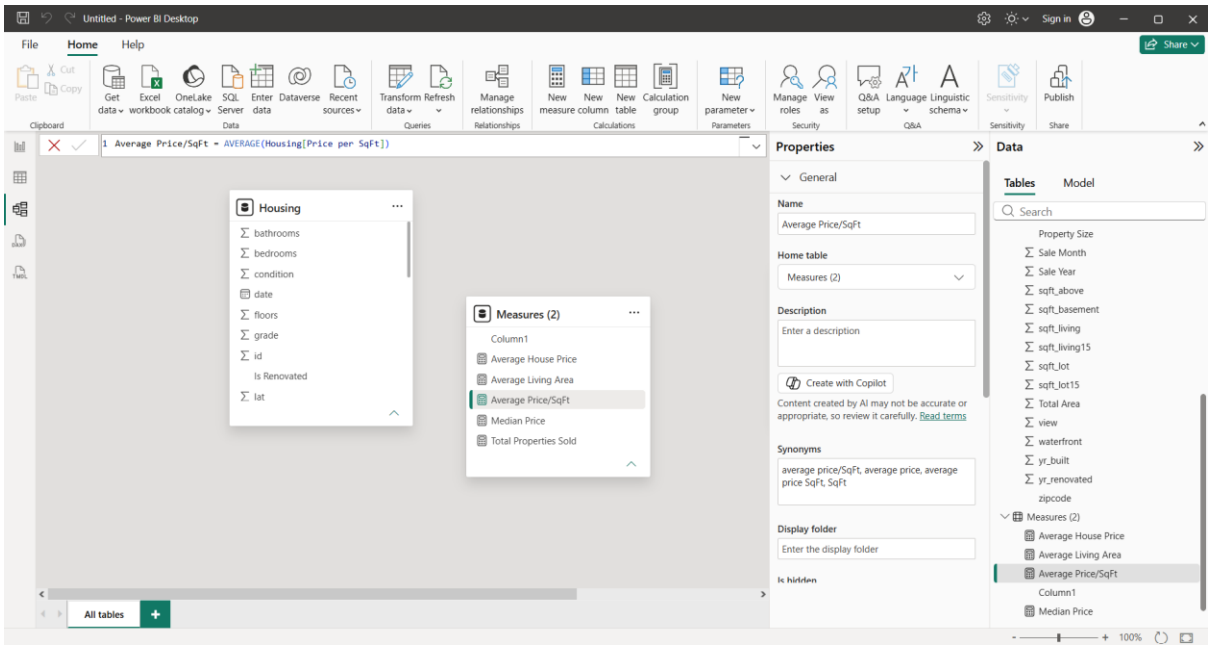
This measure calculates the middle sale price when property prices are ordered from lowest to highest, providing a central price indicator that is less affected by extreme values than the average.



The Average Living Area measure was created using the following DAX formula:

Average Living Area = AVERAGE(Housing[sqft_living])

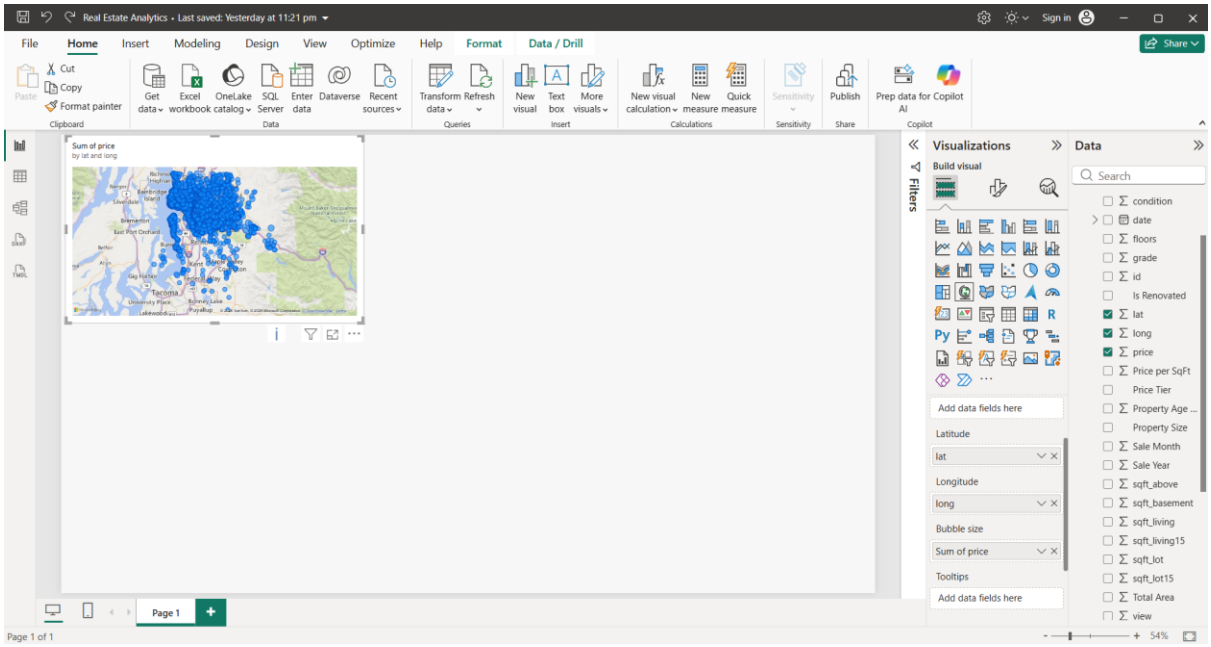
This measure calculates the average living area of the properties in square feet.



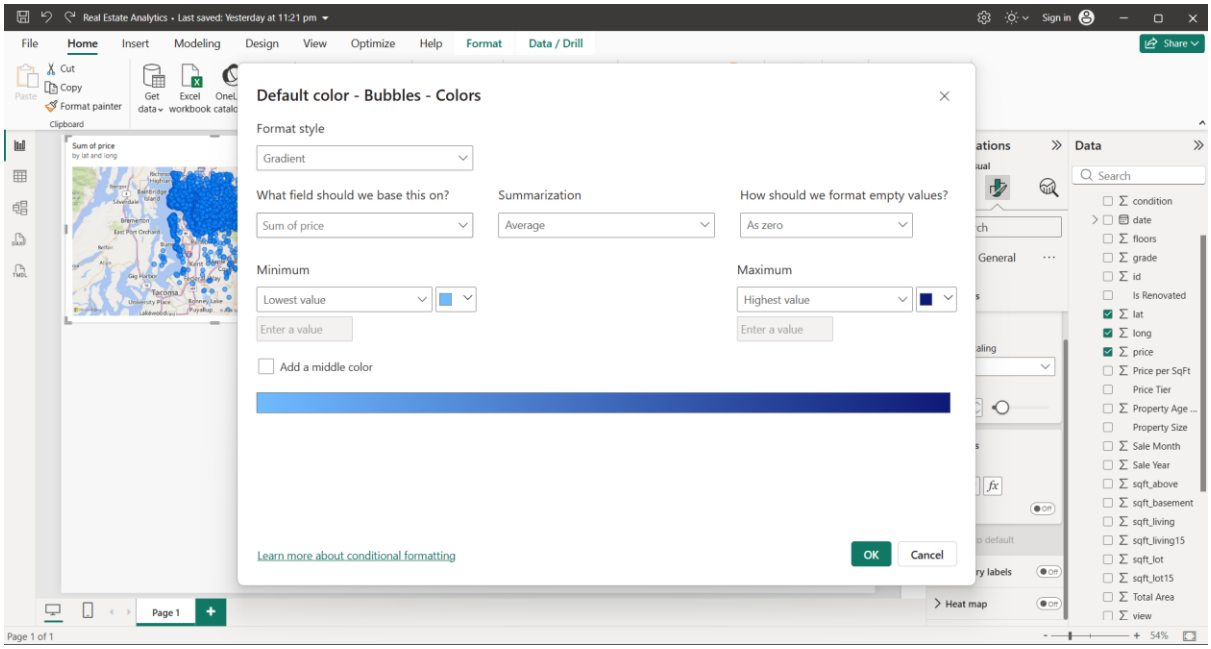
The Average Price/SqFt measure was created using the following formula:

Average Price/SqFt = AVERAGE(Housing[Price per SqFt])

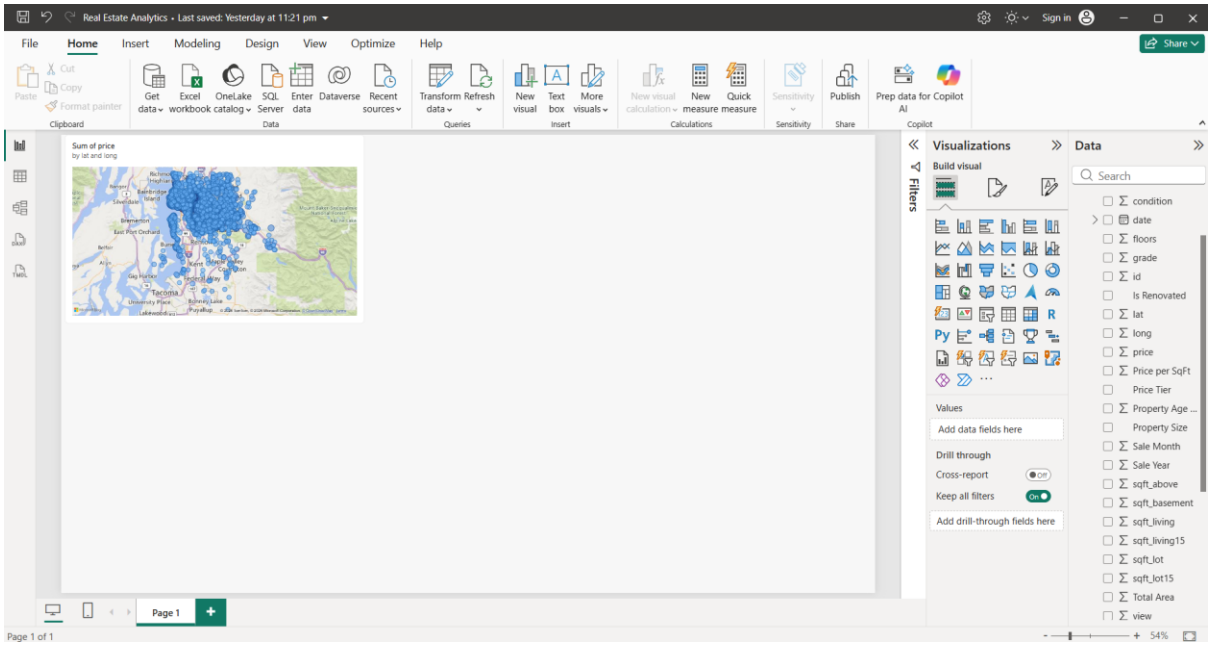
This measure calculates the average of the previously derived Price per SqFt values, enabling comparisons of property prices relative to their living area.



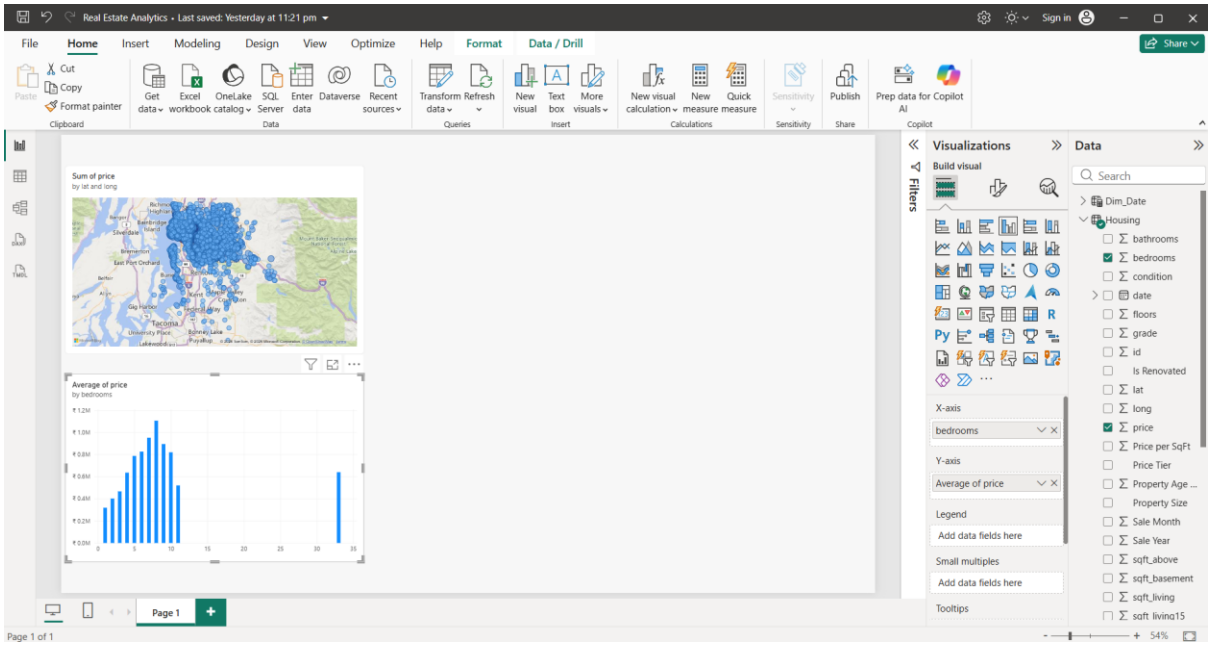
A map visual was added to the report using the latitude and longitude columns from the Housing table. The lat field was assigned to Latitude, the long field to Longitude, and Sum of price to Bubble size. This visual displays the geographic distribution of properties, with bubble sizes representing the aggregated sale price at each location.



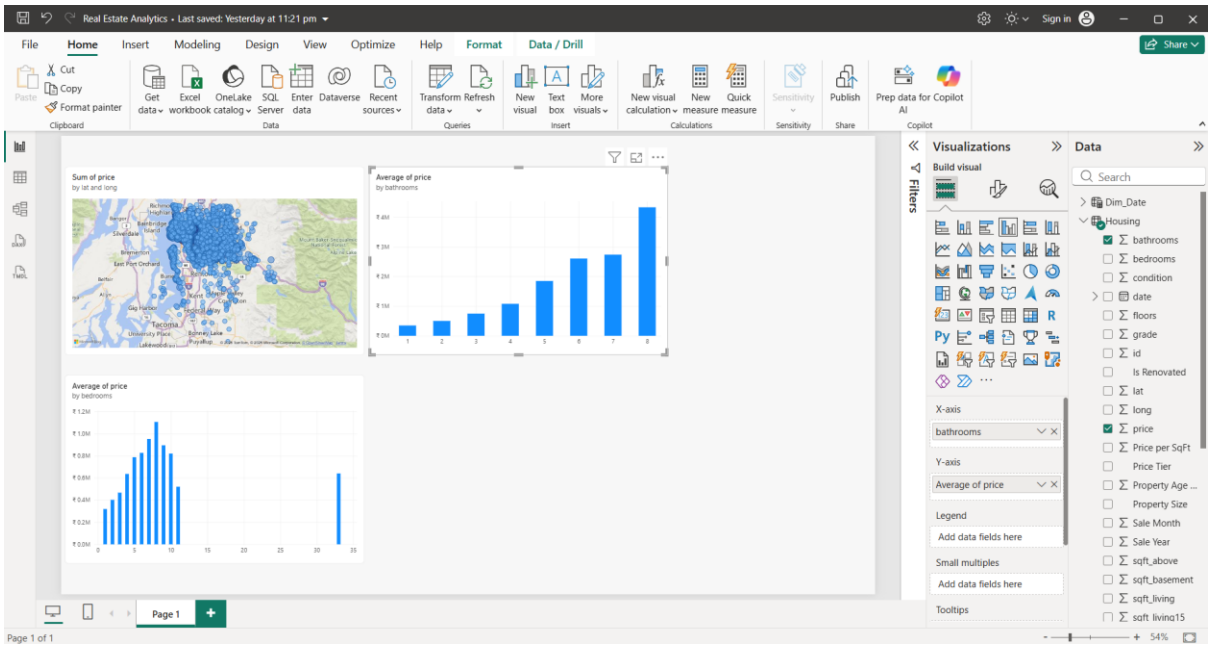
Conditional formatting was configured for the map's bubble colors using the Gradient format. The formatting was based on Sum of price, with a light-blue minimum color and a dark-blue maximum color. The summarization setting shown in the dialog was Average, and empty values were configured to display as zero.



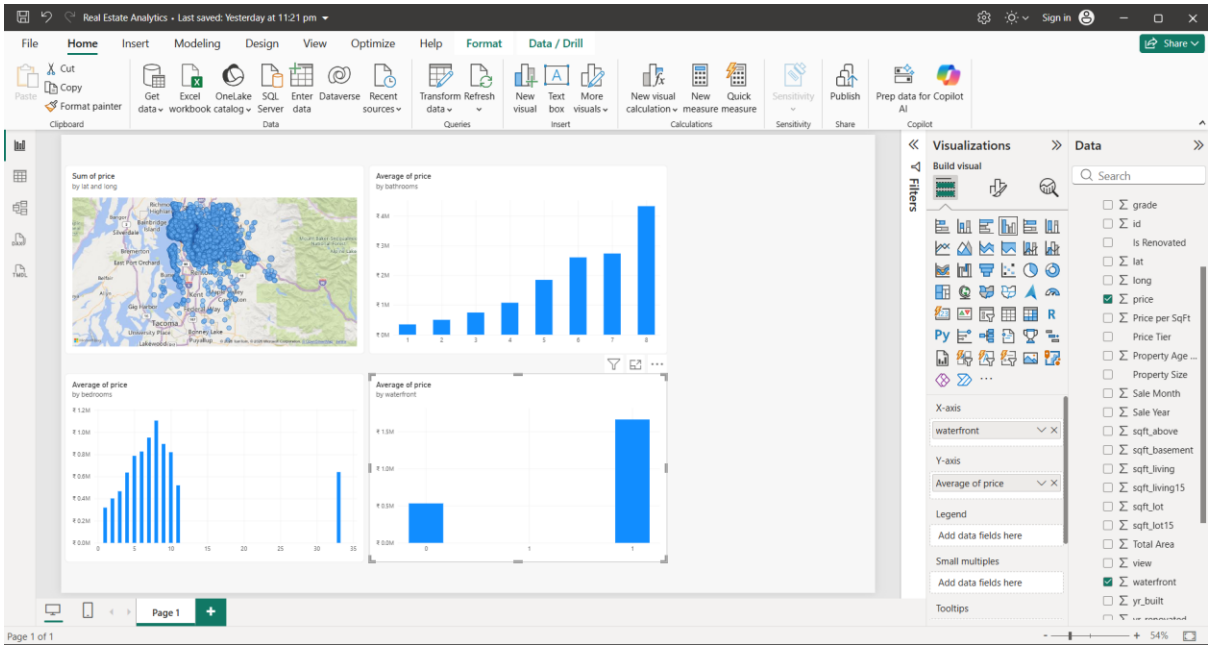
The map visual was reviewed after configuring its appearance. Property locations are displayed as blue bubbles across the geographic area, providing a spatial overview of the housing data.



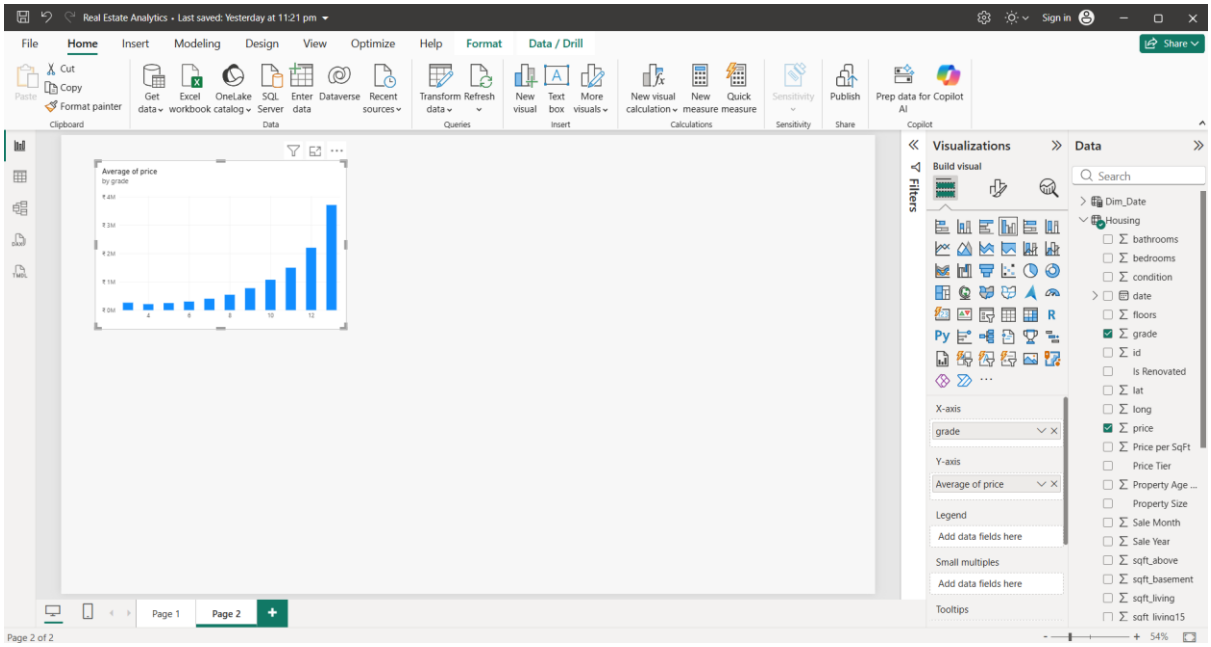
A clustered column chart was created with bedrooms on the X-axis and Average of price on the Y-axis. The chart compares average house prices across different bedroom counts and helps examine the relationship between the number of bedrooms and property prices.



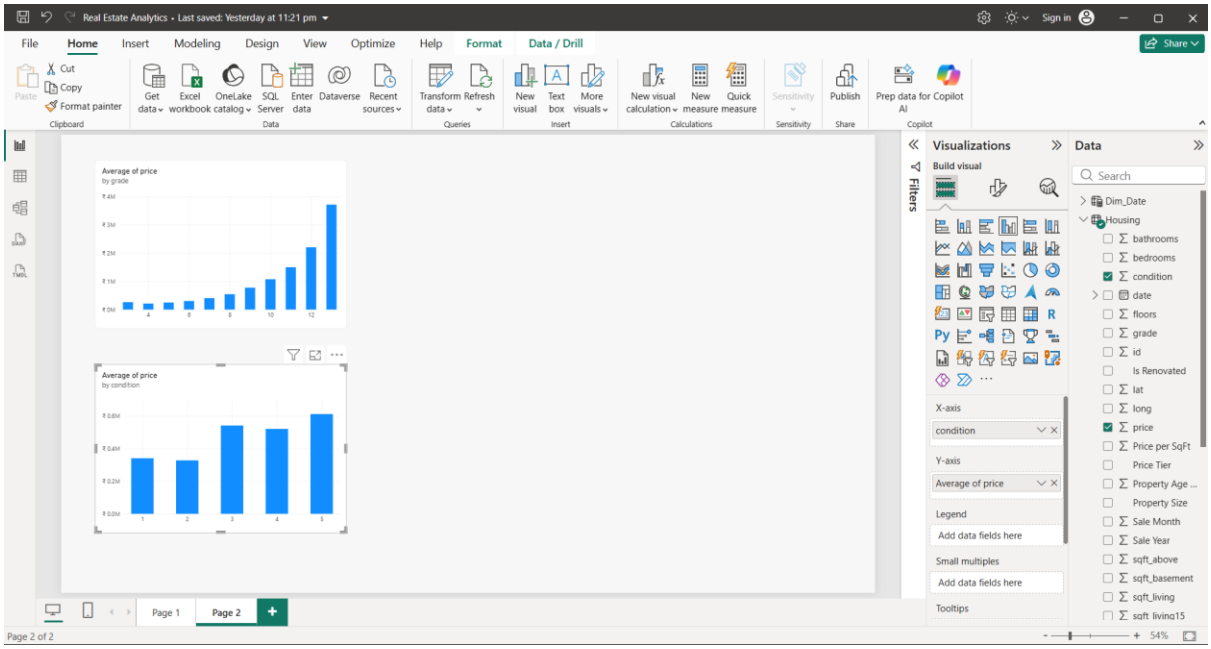
A clustered column chart was created with bathrooms on the X-axis and Average of price on the Y-axis. The visual compares average house prices across bathroom counts to examine how bathroom availability relates to property prices.



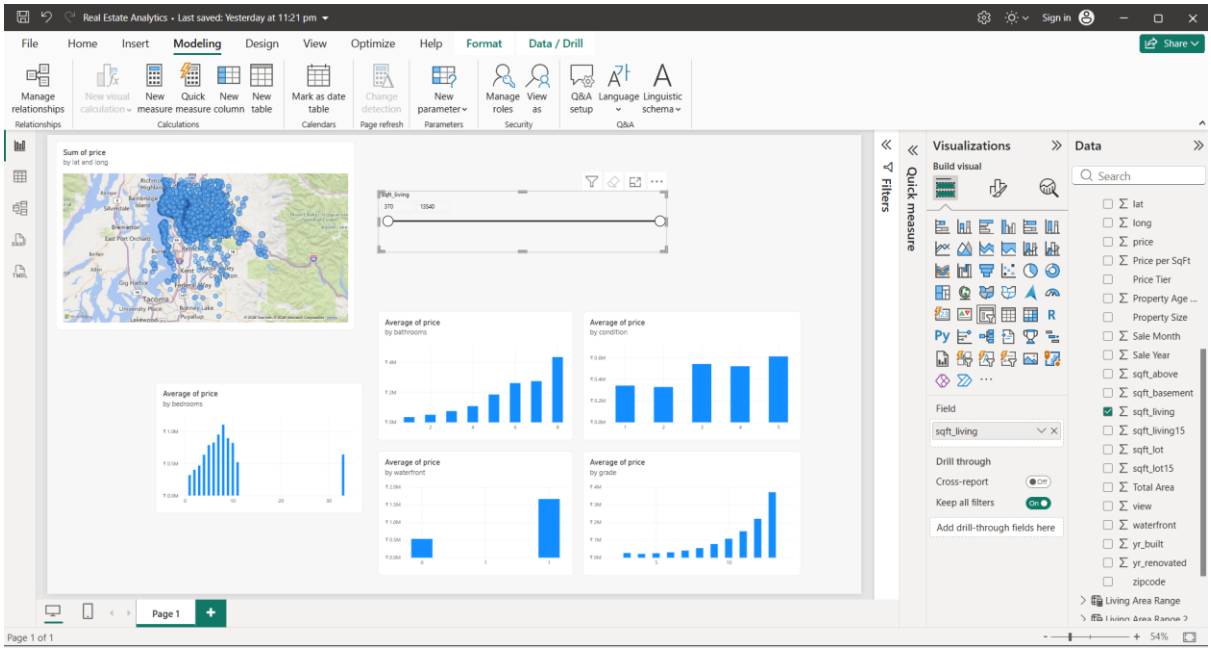
A clustered column chart was created with waterfront on the X-axis and Average of price on the Y-axis. The chart compares average prices for properties with and without waterfront access, helping assess the relationship between waterfront availability and sale prices.



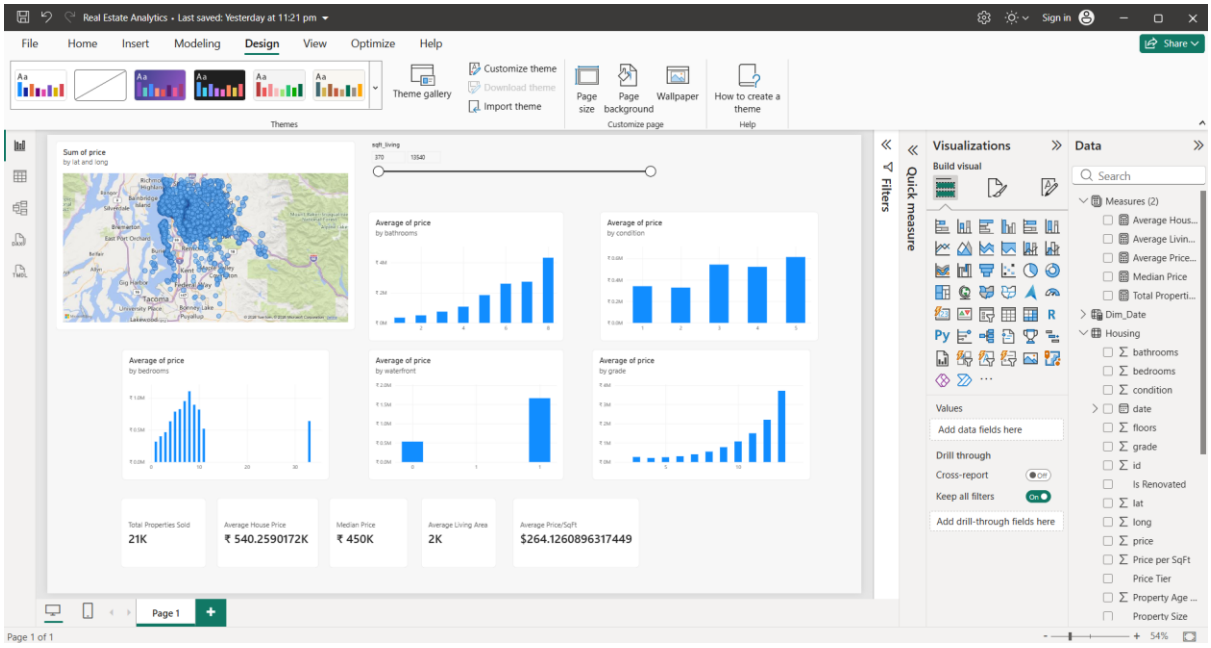
A clustered column chart was created on the second report page using grade on the X-axis and Average of price on the Y-axis. The chart compares average house prices across property grades to examine how grade classifications relate to property value.



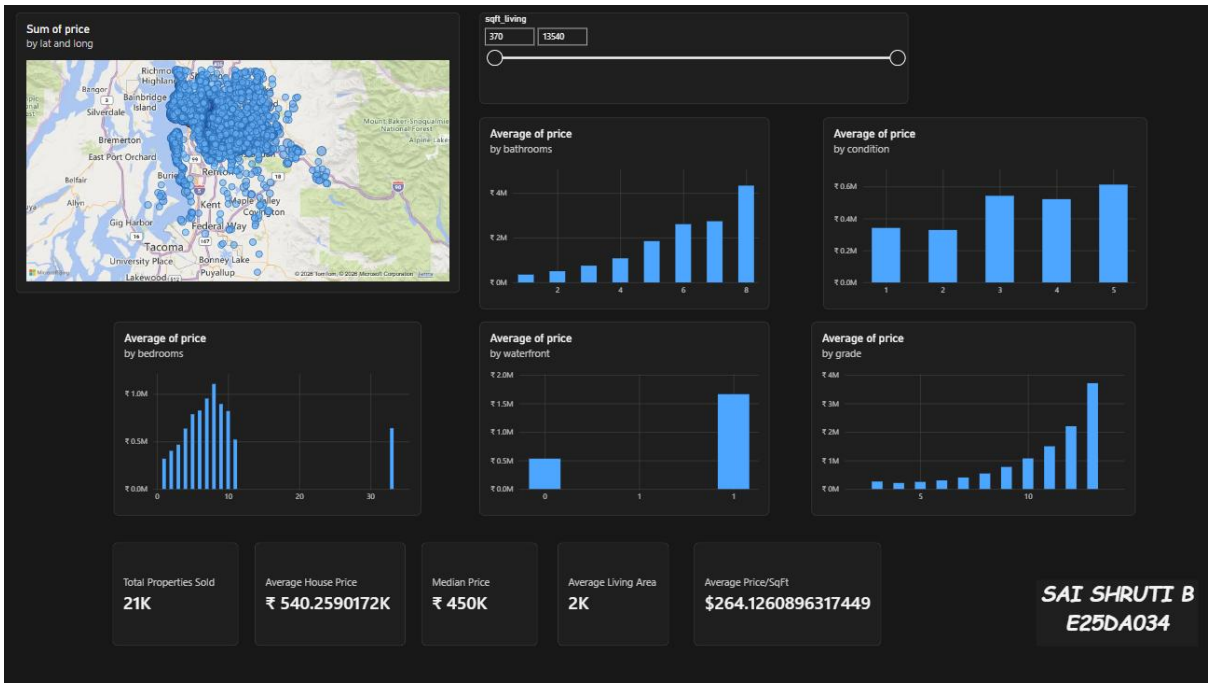
A clustered column chart was created with condition on the X-axis and Average of price on the Y-axis. The chart compares average house prices across the available condition ratings, supporting the analysis of the relationship between property condition and sale price.



A numeric Between slicer was added using the Housing[sqft_living] field. The slicer allows the report user to specify a minimum and maximum living area and filter the housing data accordingly. The map and analysis charts were arranged on the report page to support interactive exploration.



The dashboard was organized with the geographic map, living-area range slicer, and five price-analysis charts. Five card visuals were added using the required measures: Total Properties Sold, Average House Price, Median Price, Average Living Area, and Average Price/SqFt. These cards provide a concise summary of the dataset alongside the detailed charts.



The final dashboard presents the geographic distribution of properties, an interactive living-area range slicer, and charts comparing average house prices by bedrooms, bathrooms, waterfront status, grade, and condition. Five KPI

cards summarize the total number of properties, average house price, median price, average living area, and average price per square foot. A consistent dark theme and organized layout bring the visualizations together into a single real-estate analytics report.